

Questions with Answer Keys

#MathBoleTohMathonGo

Q1 - Application of LHL and RHL

The value of $\lim_{x \rightarrow \pi/2} \left[\frac{x - \frac{\pi}{2}}{\cos x} \right]$ (where, $[x]$ denotes greatest integer function) is

- (1) -1
- (2) 0
- (3) -2
- (4) Does not exist

Q2 - Application of LHL and RHL

If $f(x) = \begin{cases} x^2 + 2, & x \geq 2 \\ 1 - x, & x < 2 \end{cases}$ and $g(x) = \begin{cases} 2x, & x > 1 \\ 3 - x, & x \leq 1 \end{cases}$, then $\lim_{x \rightarrow 1} f[g(x)]$ is equal to

- (1) 4
- (2) 6
- (3) 0
- (4) Does not exist

Q3 - Application of LHL and RHL

The value of $\lim_{x \rightarrow a^-} \left(\frac{[x]^3}{a} - \left[\frac{x}{a} \right]^3 \right)$ ($a > 0$) (where, $[\cdot]$ denotes greatest integer function) is

- (1) $a^2 + 1$
- (2) a^2
- (3) $a^2 - 1$
- (4) $-a^2$

Q4 - Application of LHL and RHL

The value of $\lim_{\theta \rightarrow 0} \left(\left[\frac{n \sin \theta}{\theta} \right] + \left[\frac{n \tan \theta}{\theta} \right] \right)$, where $[\cdot]$ is greatest integer function and $n \in N$, is

- (1) $2n$
- (2) $2n + 1$
- (3) $2n - 1$

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

(4) Does not exist

Q5 - Application of LHL and RHL

Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation $(\sqrt[3]{1+a}-1)x^2 + (\sqrt{1+a}-1)x + (\sqrt[6]{1+a}-1) = 0$ where $a > -1$. Then, $\lim_{a \rightarrow 0^+} \alpha(a)$ and $\lim_{a \rightarrow 0^+} \beta(a)$ are

(1) $-\frac{5}{2}$ and 1

(2) $-\frac{1}{2}$ and -1

(3) $-\frac{7}{2}$ and 2

(4) $-\frac{9}{2}$ and 3

Q6 - Series Expansion and Conversion to Definite Integrals

The integer n for which $\lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - e^x)}{x^n}$ is a finite non-zero number, is

(1) 1

(2) 2

(3) 3

(4) 4

Q7 - Series Expansion and Conversion to Definite Integrals

The value of $\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e}{x}$ is

(1) 1

(2) $\frac{e}{2}$

(3) $-\frac{e}{2}$

(4) $\frac{2}{e}$

Q8 - Series Expansion and Conversion to Definite Integrals

The value of $\lim_{n \rightarrow \infty} \frac{[x] + [2x] + \dots + [nx]}{n^2}$ is

(1) x

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

(2) $\frac{x}{2}$

(3) $2x$

(4) $\frac{x}{4}$

Q9 - Series Expansion and Conversion to Definite Integrals

The value of $\lim_{n \rightarrow \infty} \frac{[1^2x] + [2^2x] + \dots + [n^2x]}{n^3}$ is

(1) x

(2) $\frac{x}{2}$

(3) $\frac{x}{3}$

(4) $\frac{x}{6}$

Q10 - Series Expansion and Conversion to Definite Integrals

The value of $\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2+1^2}} + \frac{1}{\sqrt{n^2+2^2}} + \dots + \frac{1}{\sqrt{n^2+n^2}} \right)$ is

(1) 1

(2) $\log(\sqrt{2})$

(3) 0

(4) $\log(1 + \sqrt{2})$

Q11 - Series Expansion and Conversion to Definite Integrals

If $f(x)$ is defined as $f : (-1, 1) \rightarrow \mathbb{R}$ and is differentiable on $(-1, 1)$. It is given that

$f'(0) = \lim_{n \rightarrow \infty} n \left[f\left(\frac{1}{n}\right) \right]$ also $f(0) = 0$. Then, the value of $\lim_{n \rightarrow \infty} \left[\frac{2}{\pi} (1+n) \cos^{-1}\left(\frac{1}{n}\right) - n \right]$ given that

$\left| \cos^{-1}\left(\frac{1}{n}\right) \right| \leq \frac{\pi}{2}$ is

(1) $\frac{1}{\pi}$

(2) $\frac{1}{\pi-2}$

(3) $\frac{\pi-2}{\pi}$

(4) π

Questions with Answer Keys

#MathBoleTohMathonGo

Q12 - Continuity

If $f(x) = |[x] - 1|$, then $f(x)$ is continuous for

- (1) $x \in R$
- (2) $x \in R - \text{Integer}$
- (3) $[0, \infty)$
- (4) $(-\infty, 0]$

Q13 - Continuity

If $f(x) = [\sin x]$, where $x \in [0, 2\pi]$, then $f(x)$ is discontinuous at x equals

- (1) $\{\pi/2, 3\pi/2\}$
- (2) $\{\pi/2, \pi, 2\pi\}$
- (3) $\{\pi, 2\pi\}$
- (4) $\{0, \pi/2, \pi, 2\pi\}$

Q14 - Continuity

If $f(x) = \min\{\tan x, \cot x\}$, then $f(x)$ is discontinuous at

- (1) $x \in n\pi$
- (2) $x \in \frac{n\pi}{2}$
- (3) $x \in (2n+1)\frac{\pi}{2}$
- (4) None of these

Q15 - Continuity

If $f(x) = \begin{cases} x+2, & \text{when } x < 1 \\ 4x-1, & \text{when } 1 \leq x \leq 3 \\ x^2+5, & \text{when } x > 3 \end{cases}$, then the correct statement is

- (1) $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 3} f(x)$
- (2) $f(x)$ is continuous at $x = 3$
- (3) $f(x)$ is continuous at $x = 1$

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

(4) $f(x)$ is continuous at $x = 1$ and 3

Q16 - Continuity

If $f(x) = \begin{cases} [x] + [-x], & x \neq 2 \\ \lambda, & x = 2 \end{cases}$ (where, $[\cdot]$ denotes greatest integer function), f is continuous at $x = 2$, then λ is equal to

(1) -1 (2) 0 (3) 1 (4) 2

Q17 - Continuity

If $f(x) = \frac{x - e^x + \cos 2x}{x^2}$, where $x \neq 0$, is continuous at $x = 0$, then (where, $\{\cdot\}$ and $[x]$ denote the fractional part and greatest integer)

(1) $f(0) = \frac{5}{2}$ (2) $[f(0)] = -2$ (3) $\{f(0)\} = -0.5$ (4) $[f(0)] \cdot \{f(x)\} = -1.5$

Q18 - Continuity

The function $f: R - \{0\} \rightarrow R$ given by $f(x) = \frac{1}{x} - \frac{2}{e^{2x}-1}$ can be made continuous at $x = 0$ defining $f(0)$ as

(1) 2 (2) -1 (3) 0 (4) 1

Q19 - Continuity

In $[1, 3]$, the function $[x^2 + 1]$, $[\cdot]$ denote the greatest integer function, is continuous

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

(1) for all x (2) for all x except at nine points(3) for all x except at seven points(4) for all x except at eight points

Q20 - Continuity

The number of points of discontinuity of $f(x) = [2x^3 - 5]$ in $[1, 2)$ is equal to (where, $[\cdot]$ denotes greatest integer function)

(1) 14

(2) 13

(3) 10

(4) 8

Q21 - Continuity

Given, $f(x) = \begin{cases} |x+1|, & \text{if } x < -2 \\ 2x+3, & \text{if } -2 \leq x < 0 \\ x^2+3, & \text{if } 0 \leq x < 3 \\ x^3-15, & \text{if } 3 \leq x \end{cases}$, then the number of points of discontinuity of $f(x)$ is

(1) 0

(2) 1

(3) 2

(4) 3

Q22 - Continuity

Let $f: R \rightarrow R$ be a continuous function, $\forall x \in R$ and $f(x) = 5, \forall x \in \text{irrational}$. Then, the value of $f(3)$ is

(1) 1

(2) 2

(3) 5

Questions with Answer Keys

#MathBoleTohMathonGo

(4) Cannot be determined

Q23 - Continuity

Let $f(x)$ be a continuous function which satisfies $f(x^2 + 1) = \frac{x^2}{f(2^x) - 1}$ and $f(x) > 0$ for all $x \in R$, then

$\lim_{x \rightarrow 1} f(x)$ is equal to

(1) 2

(2) 4

(3) 1

(4) Does not exist

Q24 - Continuity

Let function f be defined as $f: R^+ \rightarrow R^+$ and function g is defined as $g: R^+ \rightarrow R^+$. Functions f and g are continuous in their domain. Suppose, the function $h(x) = \lim_{n \rightarrow \infty} \frac{x^n f(x) + x^2}{x^n + g(x)}$, $x > 0$. If $h(x)$ is continuous in its domain, then $f(1) \cdot g(1)$ is equal to

(1) 2

(2) 1

(3) $1/2$

(4) 0

Q25 - Differentiability

If $f(x) = x^3 \operatorname{Sgn}(x)$, then

(1) f is differentiable at $x = 0$ (2) f is continuous but not differentiable at $x = 0$ (3) $f'(0^-) = 1$

(4) None of the above

Q26 - Differentiability

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

The function $f(x) = (x^2 - 1)|x^2 - 3x + 2| + \cos(|x|)$ is not differentiable at $x =$

(1) -1

(2) 0

(3) 1

(4) 2

Q27 - Differentiability

If $f(x) = \begin{cases} xe^{-(\frac{1}{|x|} + \frac{1}{x})}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, then $f(x)$ is

(1) discontinuous everywhere

(2) continuous as well as differentiable at all x

(3) continuous for all x but not differentiable at $x = 0$

(4) neither differentiable nor continuous at $x = 0$

Q28 - Differentiability

The left hand derivative of $f(x) = [x] \sin(\pi x)$ at $x = k$, k is an integer is (where, $[\cdot]$ denotes greatest integer function)

(1) $(-1)^k \cdot (k-1)\pi$

(2) $(-1)^{k-1} \cdot (k-1)\pi$

(3) $(-1)^k \cdot k\pi$

(4) $(-1)^{k-1} \cdot k\pi$

Q29 - Differentiability

The function given by $y = ||x| - 1|$ is differentiable for all real numbers except the points

(1) $\{0, 1, -1\}$

(2) ± 1

(3) 1

Questions with Answer Keys

#MathBoleTohMathonGo

(4) -1

Q30 - Differentiability

If $f(x) = \begin{cases} x + \{x\} + x \sin\{x\}, & \text{for } x \neq 0 \\ 0, & \text{for } x = 0 \end{cases}$ where $\{\cdot\}$ denotes fractional part of x , then

- (1) f is continuous and differentiable at $x = 0$
- (2) f is continuous but not differentiable at $x = 0$
- (3) f is neither continuous nor differentiable at $x = 0$
- (4) None of the above

Q31 - Differentiability

Let $f(x) = x^3$ and $g(x) = |x|$, then at $x = 0$, the composite function

- (1) $g \circ f$ is derivable but $f \circ g$ is not
- (2) $f \circ g$ is derivable but $g \circ f$ is not
- (3) $g \circ f$ and $f \circ g$ both are derivable
- (4) neither $g \circ f$ nor $f \circ g$ is derivable

Q32 - Differentiability

Identify the correct statement.

- (1) If $f(x)$ is derivable at $x = a$, $|f(x)|$ will also be derivable at $x = a$
- (2) If $f(x)$ is continuous at $x = a$, $|f(x)|$ too will be continuous at $x = a$
- (3) If $f(x)$ is discontinuous at $x = a$, $|f(x)|$ will also be discontinuous at $x = a$
- (4) If $|f(x)|$ is continuous at $x = a$, $f(x)$ too will be continuous at $x = a$

Q33 - Differentiability

A function f defined as $f(x) = x[x]$ for $-1 \leq x \leq 3$ (where, $[\cdot]$ denotes greatest integer function) is

- (1) continuous at all points in the domain of f but non-derivable at a finite number of points

Questions with Answer Keys

#MathBoleTohMathonGo

(2) discontinuous at all points and hence non-derivable at all points in the domain of f

(3) discontinuous at a finite number of points but not derivable at all points in the domain of f

(4) discontinuous and also non-derivable at a finite number of point of f

Q34 - Differentiability

Let the function f, g and h be defined as follows:

$$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & \text{for } -1 \leq x \leq 1 \text{ and } x \neq 0 \\ 0, & \text{for } x = 0 \end{cases}$$

$$g(x) = \begin{cases} x^2 \cdot \sin\left(\frac{1}{x}\right), & \text{for } -1 \leq x \leq 1 \text{ and } x \neq 0 \\ 0, & \text{for } x = 0 \end{cases}$$

$$h(x) = |x|^3, \text{ for } -1 \leq x \leq 1$$

Which of these functions are differentiable at $x = 0$?

(1) f and g

(2) f and h

(3) g and h

(4) None of these

Q35 - Differentiability

The domain of derivative of the function $f(x) = \begin{cases} \tan^{-1} x, & \text{if } |x| \leq 1 \\ \frac{1}{2}(|x| - 1), & \text{if } |x| > 1 \end{cases}$, is

(1) $\mathbb{R} - \{0\}$

(2) $\mathbb{R} - \{1\}$

(3) $\mathbb{R} - \{-1\}$

(4) $\mathbb{R} - \{-1, 1\}$

Questions with Answer Keys

#MathBoleTohMathonGo

Answer Key

Q1 (3)	Q2 (2)	Q3 (2)	Q4 (3)
Q5 (2)	Q6 (3)	Q7 (3)	Q8 (2)
Q9 (3)	Q10 (4)	Q11 (3)	Q12 (2)
Q13 (2)	Q14 (2)	Q15 (3)	Q16 (1)
Q17 (4)	Q18 (4)	Q19 (4)	Q20 (2)
Q21 (2)	Q22 (3)	Q23 (1)	Q24 (2)
Q25 (1)	Q26 (4)	Q27 (3)	Q28 (1)
Q29 (1)	Q30 (3)	Q31 (3)	Q32 (2)
Q33 (4)	Q34 (3)	Q35 (4)	
